

# New Keynesian Models: Complete Study Notes

Based on Romer, *Advanced Macroeconomics*, Chapters 6–7  
Classes 8 and 9

Carlos

## Contents

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|          |                                                                     |           |
|----------|---------------------------------------------------------------------|-----------|
| <b>1</b> | <b>1. Starting Point: Real Business Cycles (RBC)</b>                | <b>3</b>  |
| 1.1      | Core Idea . . . . .                                                 | 3         |
| 1.2      | The Classical Dichotomy . . . . .                                   | 3         |
| <b>2</b> | <b>2. Motivation for New Keynesian Models</b>                       | <b>4</b>  |
| 2.1      | Incomplete / Sluggish Nominal Adjustment . . . . .                  | 4         |
| 2.2      | Real Shocks and a New Channel . . . . .                             | 4         |
| <b>3</b> | <b>3. A Model of Imperfect Competition and Menu Costs (Class 8)</b> | <b>4</b>  |
| 3.1      | Setting . . . . .                                                   | 4         |
| 3.2      | The CES Consumption Index . . . . .                                 | 5         |
| 3.3      | Aggregate Demand . . . . .                                          | 5         |
| <b>4</b> | <b>4. The Household's Problem (Static Model)</b>                    | <b>6</b>  |
| 4.1      | Stage 1: Allocating the Budget Among Goods . . . . .                | 6         |
| 4.2      | Stage 2: Choosing How Much to Work . . . . .                        | 7         |
| <b>5</b> | <b>5. The Firm's Problem (Static Model)</b>                         | <b>7</b>  |
| <b>6</b> | <b>6. Equilibrium (Static Model)</b>                                | <b>8</b>  |
| 6.1      | From Flexible Prices to the Optimal Price . . . . .                 | 8         |
| 6.2      | Symmetric Equilibrium . . . . .                                     | 9         |
| 6.3      | Aggregate Demand and the Optimal Price in Logs . . . . .            | 9         |
| <b>7</b> | <b>7. Adding Price-Adjustment Costs</b>                             | <b>9</b>  |
| 7.1      | The Firm's Incentive to Adjust . . . . .                            | 9         |
| 7.2      | Two Cases . . . . .                                                 | 10        |
| 7.3      | The Key Result . . . . .                                            | 10        |
| 7.4      | Implications and the Aggregate Demand Externality . . . . .         | 11        |
| 7.5      | Real Rigidity . . . . .                                             | 11        |
| <b>8</b> | <b>8. Dynamic New Keynesian Models (Class 9)</b>                    | <b>11</b> |

|           |                                                             |           |
|-----------|-------------------------------------------------------------|-----------|
| <b>9</b>  | <b>9. Households (Dynamic Model)</b>                        | <b>12</b> |
| 9.1       | Objective Function . . . . .                                | 12        |
| 9.2       | Labor Supply FOC . . . . .                                  | 12        |
| 9.3       | The Euler Equation and the New Keynesian IS Curve . . . . . | 13        |
| <b>10</b> | <b>10. Firms' Problem (Dynamic Model)</b>                   | <b>14</b> |
| 10.1      | Setup . . . . .                                             | 14        |
| 10.2      | The Firm's Maximization Problem . . . . .                   | 15        |
| 10.3      | Simplification . . . . .                                    | 15        |
| 10.4      | FOC and Solution . . . . .                                  | 15        |
| 10.5      | The New Keynesian Aggregate Supply Curve . . . . .          | 16        |
| <b>11</b> | <b>11. Summary and Connection</b>                           | <b>17</b> |

## 1. Starting Point: Real Business Cycles (RBC)

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### Core Idea

The RBC model combines a stochastic growth model with calibration to generate quantitative implications for business cycles. Booms and recessions are *optimal responses* of the economy to technology (productivity) shocks. There is nothing to fix — fluctuations are efficient.

#### Key assumptions:

- Prices (goods prices and wages) adjust **instantaneously** to clear all markets.
- Markets are perfectly competitive — all agents take prices as given.
- Aggregate supply is **vertical**.

#### What is aggregate supply?

Aggregate supply (oferta agregada) is the total quantity of goods and services that all firms in the economy are willing to produce. “Aggregate” means we sum over all firms in the country.

### The Classical Dichotomy

Because aggregate supply is vertical, nominal disturbances (e.g. monetary policy) only affect the *nominal* side of the economy — prices, wages, money — but not *real* variables such as output and employment.

#### What does “nominal” mean?

**Nominal** = expressed in money terms — the number on the price tag.

**Real** = how much you can actually buy with that money.

If the central bank doubles the money supply and aggregate supply is vertical, firms cannot produce more — they can only raise prices proportionally. Real output is unchanged. Money is **neutral**.

#### Is $Y$ always constant in RBC?

No.  $Y$  changes, but only due to **real** shocks (technology, labor supply). What is constant is  $Y$ 's response to nominal disturbances — monetary policy cannot move real output in RBC.

**Classical Dichotomy:** Aggregate supply is vertical  $\Rightarrow$  nominal disturbances only affect the nominal side of the economy. Money is neutral.

## 2. Motivation for New Keynesian Models

### Incomplete / Sluggish Nominal Adjustment

In reality, prices and wages do **not** adjust instantaneously. This is called **nominal rigidity**.

#### What is nominal rigidity?

Prices or wages cannot easily or quickly adjust. When prices are sticky and demand rises, firms *produce more* rather than simply raising prices. This is the key channel through which monetary disturbances have real effects.

Incomplete nominal adjustment  $\Rightarrow$  changes in aggregate demand affect output and employment

$\Rightarrow$  Nominal disturbances cause changes in employment and output (at least in the short run)

### Real Shocks and a New Channel

Real shocks (government spending, oil prices, etc.) also affect aggregate demand at a given price level. With incomplete price adjustment, this creates an **additional channel**: shocks now affect output and employment more strongly, because prices cannot immediately absorb them.

#### RBC vs. New Keynesian

|                  | RBC              | New Keynesian           |
|------------------|------------------|-------------------------|
| Prices           | Fully flexible   | Sticky                  |
| Money            | Neutral          | Non-neutral (short run) |
| What moves $Y$ ? | Only real shocks | Real and nominal shocks |
| Time horizon     | —                | Short run vs. long run  |

## 3. A Model of Imperfect Competition and Menu Costs (Class 8)

### Setting

Production function of firm  $i$ :

$$Y_i = L_i$$

One unit of labor produces one unit of output.

**Household utility:**

$$U = C - \frac{1}{\gamma} L^\gamma, \quad \gamma > 1$$

- $C$ : consumption index — more consumption, more utility.
- $-\frac{1}{\gamma} L^\gamma$ : disutility of working — working hurts (negative sign).
- $\gamma > 1$ : controls how fast the pain of working grows.

### What is disutility?

Disutility is simply how much you dislike something. Working costs effort and time — you prefer not to do it. That cost to your wellbeing is the disutility of labor. It appears with a negative sign in utility because it subtracts from your total welfare.

### Why does $\gamma$ affect how much you work?

From the optimal labor supply  $L = (W/P)^{1/(\gamma-1)}$ , a higher  $\gamma$  reduces  $L$  because each extra hour of work becomes increasingly painful.  $\gamma$  also controls the elasticity of labor supply  $1/(\gamma - 1)$ : how sensitive your hours are to the real wage.

## The CES Consumption Index

$$C = \left[ \int_{i=0}^1 C_i^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}}, \quad \eta > 1$$

$C_i$  = quantity consumed of good  $i$ . The integral sums over all goods.  $\eta$  = elasticity of substitution: how easily consumers switch between goods.

- High  $\eta$ : goods are similar — if a firm raises its price, it loses almost all customers.
- Low  $\eta$ : goods are differentiated — firms have more pricing power.

**Simplifications:** No capital, no government, no trade  $\Rightarrow Y = C$ .

## Aggregate Demand

$$Y = \frac{M}{P}$$

Three microfoundations:

1. **Real money balances in the utility function:** Money provides convenience. Optimizing yields  $Y = M/P$ .
2. **Cash-in-advance constraint:**  $PC \leq M \Rightarrow Y = M/P$ .
3. **Central bank targets nominal GDP:**  $PY = M \Rightarrow Y = M/P$ .

### If $M$ rises, doesn't $P$ rise too, leaving $Y$ unchanged?

With flexible prices (RBC): yes — classical dichotomy holds.

With sticky prices (New Keynesian):  $P$  cannot rise immediately, so  $Y$  rises in the short run. The whole model lives in this short run.

### What is imperfect competition? What are menu costs?

**Imperfect competition:** Firms have market power and *choose* their own price. This is necessary for price rigidity to matter — in perfect competition the market dictates prices, so there is nothing to keep fixed.

**Menu costs:** The cost of *changing* a price (reprinting catalogues, reprogramming

systems, renegotiating contracts). If the cost of changing exceeds the gain, the firm keeps its price fixed. This generates nominal rigidity naturally.

## 4. The Household's Problem (Static Model)

### Stage 1: Allocating the Budget Among Goods

$$\mathcal{L} = \left[ \int_0^1 C_i^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}} + \lambda \left( S - \int_0^1 P_i C_i di \right)$$

- $P_i$ : price of good  $i$
- $C_i$ : quantity consumed of good  $i$
- $P_i C_i$ : spending on good  $i$
- $S$ : total budget
- $\lambda$ : Lagrange multiplier — how much utility rises with one more unit of budget

**Why does this Lagrangian not include the full utility  $U = C - \frac{1}{\gamma} L^\gamma$ ?**

The problem is split into two stages. In Stage 1,  $L$  is fixed. Since  $-\frac{1}{\gamma} L^\gamma$  is then a constant, maximizing  $U$  is identical to maximizing  $C$ . The disutility of labor is handled in Stage 2.

### Results from solving the Lagrangian:

$$C = \frac{S}{P} \quad (\text{real consumption} = \text{real budget})$$

$$C_i = \left( \frac{P_i}{P} \right)^{-\eta} C \quad (\text{demand for good } i)$$

### Why the $-\eta$ exponent?

$(P_i/P)^{-\eta} = 1/(P_i/P)^\eta$ . When  $P_i/P$  rises (good  $i$  becomes relatively expensive),  $C_i$  falls. The negative exponent captures the inverse relationship between relative price and demand. The size of  $\eta$  determines how strongly demand responds: high  $\eta$  = very sensitive to price differences.

### What is $P_i/P$ ?

$P_i$  is the nominal price of good  $i$  in pesos.  $P$  is the general price level — the average price of all goods.  $P_i/P$  is the *relative price* of good  $i$ : how expensive it is compared to the average.

$P_i/P = 1$ : your good costs exactly the average.

$P_i/P > 1$ : your good is more expensive than average.

The real wage  $W/P$  follows the same logic:  $W$  in pesos, divided by  $P$ , gives how much you can actually buy.

## Stage 2: Choosing How Much to Work

Total income:  $S = WL + R$ , where  $WL$  is labor income and  $R$  is profit income.

### What is $R$ ?

$R$  is the firm's profit, distributed back to the household as the owner. In this model, households own all firms, so profits return to them as dividend income. Total household income therefore has two components: labor income  $WL$  and profit income  $R$ .

$$\max_L \frac{WL + R}{P} - \frac{1}{\gamma} L^\gamma$$

FOC for  $L$ :

$$\frac{W}{P} - L^{\gamma-1} = 0 \quad \Rightarrow \quad L = \left( \frac{W}{P} \right)^{\frac{1}{\gamma-1}}$$

### Intuition for the FOC:

The FOC equates marginal benefit and marginal cost of working:

**Marginal benefit** =  $W/P$  = real wage (one extra hour gives  $W/P$  more real consumption).

**Marginal cost** =  $L^{\gamma-1}$  = marginal disutility of labor.

Labor supply is an increasing function of the real wage  $W/P$ : if  $W/P$  is high, working is worth it; if low, it is not.

Labor supply elasticity =  $\frac{1}{\gamma-1}$ .

## 5. The Firm's Problem (Static Model)

The firm maximizes real profits:

$$\frac{R_i}{P} = \underbrace{\frac{P_i}{P} Y_i}_{\text{real revenue}} - \underbrace{\frac{W}{P} L_i}_{\text{real cost}}$$

Substituting  $Y_i = L_i$ ,  $C_i = (P_i/P)^{-\eta} Y$ , and  $C_i = Y_i$ :

$$\frac{R_i}{P} = \left( \frac{P_i}{P} \right)^{1-\eta} Y - \frac{W}{P} \left( \frac{P_i}{P} \right)^{-\eta} Y$$

### Derivation: factoring the profit function

Factor out  $\left( \frac{P_i}{P} \right)^{-\eta} Y$ :

$$\frac{R_i}{P} = \left( \frac{P_i}{P} \right)^{-\eta} Y \left( \frac{P_i}{P} - \frac{W}{P} \right)$$

This is: quantity sold  $\times$  (real price  $-$  real cost). Intuitive.

**FOC for  $P_i/P$ :**

$$(1 - \eta) \left( \frac{P_i}{P} \right)^{-\eta} Y + \eta \frac{W}{P} \left( \frac{P_i}{P} \right)^{-\eta-1} Y = 0$$
$$\Rightarrow \boxed{\frac{P_i}{P} = \frac{\eta}{\eta - 1} \cdot \frac{W}{P}}$$

### What is a markup?

A markup is the factor by which price exceeds marginal cost. Here, the firm's only cost is  $W/P$ , and the markup is  $\frac{\eta}{\eta-1} > 1$  always (since  $\eta > 1$ ).

| $\eta$               | Situation           | Markup                 |
|----------------------|---------------------|------------------------|
| 2                    | Few substitutes     | $2/1 = 2$              |
| 5                    | Some substitutes    | $5/4 = 1.25$           |
| 10                   | Many substitutes    | $10/9 \approx 1.11$    |
| $\rightarrow \infty$ | Perfect competition | Markup $\rightarrow 1$ |

### Can the markup be negative?

No. Since  $\eta > 1$  always, both numerator and denominator of  $\frac{\eta}{\eta-1}$  are positive, so the markup is always  $> 1$ . A markup below 1 would mean selling below cost — no optimizing firm does that voluntarily. The assumption  $\eta > 1$  also ensures a finite optimal price exists.

## 6. Equilibrium (Static Model)

### From Flexible Prices to the Optimal Price

#### Step 1: Express $W/P$ in terms of $Y$

From the household's FOC:  $L = (W/P)^{1/(\gamma-1)}$ . In equilibrium  $Y = L$ , so:

$$Y = \left( \frac{W}{P} \right)^{\frac{1}{\gamma-1}} \Rightarrow \frac{W}{P} = Y^{\gamma-1}$$

Higher output requires a higher real wage to induce households to supply that much labor.

#### Step 2: Substitute into the firm's pricing rule

$$\frac{P_i^*}{P} = \frac{\eta}{\eta - 1} \cdot Y^{\gamma-1}$$

#### Step 3: Take logarithms

$$\ln \left( \frac{P_i^*}{P} \right) = \ln \frac{\eta}{\eta - 1} + \ln Y^{\gamma-1}$$



Using  $\ln(P_i^*/P) = p_i^* - p$  and  $\ln Y^{\gamma-1} = (\gamma-1)y$ :

$$p_i^* - p = \underbrace{\ln \frac{\eta}{\eta-1}}_c + \underbrace{(\gamma-1)}_\phi y \equiv c + \phi y$$

## Symmetric Equilibrium

Since all firms are identical, they all set  $P_i = P$  for all  $i$ , so  $P_i^*/P = 1$ :

### Solving for equilibrium output

$$\frac{\eta}{\eta-1} Y^{\gamma-1} = 1 \quad \Rightarrow \quad Y^{\gamma-1} = \frac{\eta-1}{\eta}$$

$$Y = \left( \frac{\eta-1}{\eta} \right)^{\frac{1}{\gamma-1}} < 1$$

$Y < 1$  because  $(\eta-1)/\eta < 1$ . This reflects monopolistic inefficiency: firms charge a markup and produce less than the socially optimal level. With perfect competition ( $\eta \rightarrow \infty$ ),  $Y \rightarrow 1$ .

## Aggregate Demand and the Optimal Price in Logs

**Deriving**  $p_i^* = \phi m_t + (1-\phi)p_t$

The aggregate demand equation  $Y = M/P$  in logs:

$$y = m - p \quad \Rightarrow \quad p = m - y$$

Substituting into  $p_i^* = p + c + \phi y$ :

$$p_i^* = (m - y) + c + \phi y = c + m + (\phi - 1)y$$

$$p_i^* = c + (1-\phi)p + \phi m$$

The optimal price depends on the general price level  $p$  and the money supply  $m$ . More money  $\Rightarrow$  more demand  $\Rightarrow$  higher optimal price.

## 7. Adding Price-Adjustment Costs

### The Firm's Incentive to Adjust

Let  $v = \frac{1}{\gamma-1}$  (elasticity of labor supply). The profit function is:

$$\Pi_i = Y \left( \frac{P_i}{P} \right)^{-\eta} \left( \frac{P_i}{P} - Y^{1/v} \right)$$

Substituting  $Y = M/P$ :

$$\Pi_i = \frac{M}{P} \left( \frac{P_i}{P} \right)^{1-\eta} - \left( \frac{M}{P} \right)^{\frac{1+v}{v}} \left( \frac{P_i}{P} \right)^{-\eta}$$

**Where does  $Y^{1/v}$  come from?**

We had  $W/P = Y^{\gamma-1}$ . Since  $v = 1/(\gamma - 1)$ , we have  $\gamma - 1 = 1/v$ , so  $W/P = Y^{1/v}$ . Same expression, new notation.

## Two Cases

With parameters  $v = 0.1$ ,  $\eta = 5$  (markup 25%), equilibrium output  $Y^{eq} = 0.978$ . Now consider a 3% fall in  $M$ .

**Case 1: Firm does NOT adjust**  $\rightarrow P_i = P$

Substitute  $P_i/P = 1$ :

$$\Pi_{fixed} = \frac{M}{P} (1)^{1-\eta} - \left( \frac{M}{P} \right)^{\frac{1+v}{v}} (1)^{-\eta} = \boxed{\frac{M}{P} - \left( \frac{M}{P} \right)^{\frac{1+v}{v}}}$$

**Case 2: Firm DOES adjust**  $\rightarrow P_i = P_i^*$

The new optimal price is  $P_i^* = \frac{\eta}{\eta-1} \left( \frac{M}{P} \right)^{1/v}$ .

**Where does  $\left( \frac{M}{P} \right)^{1/v}$  come from?**

Chain of substitutions:

$$\frac{P_i^*}{P} = \frac{\eta}{\eta-1} \cdot \frac{W}{P} = \frac{\eta}{\eta-1} \cdot Y^{1/v} = \frac{\eta}{\eta-1} \cdot \left( \frac{M}{P} \right)^{1/v}$$

After substituting and simplifying:

$$\Pi_{adj} = \frac{1}{\eta-1} \left( \frac{\eta}{\eta-1} \right)^{-\eta} \left( \frac{M}{P} \right)^{\frac{1+v-\eta}{v}}$$

## The Key Result

$$\Pi_{adj} - \Pi_{fixed} = 0.253\%$$

The private incentive to adjust is only 0.253% of revenue — extremely small.

**Why does this result matter?**

Since the gain from adjusting is so small, virtually any menu cost — however tiny — will exceed it, and the firm rationally chooses **not to adjust**.

This is nominal rigidity: not a physical inability to change prices, but a rational decision not to, because the private gain is too small to justify the cost.

**But wait — this means money IS neutral?**

At first glance, yes. But the deeper point is that for menu costs to generate significant rigidity, we need additional mechanisms: **real rigidity** and/or an **insensitive profit**

**function.** See below.

### Implications and the Aggregate Demand Externality

- The gap between equilibrium and optimal output is greater when  $\eta$  is lower (more market power) and when  $\gamma$  is lower (more elastic labor supply).
- Pricing decisions have **externalities** operating through aggregate demand.

#### What is the aggregate demand externality?

When one firm does not adjust its price, it keeps its output higher, which sustains aggregate demand  $Y = M/P$  (since  $P$  rises less). This benefits all other firms, who also sell more. The firm that did not adjust generated a positive externality for all others — without intending to. Since each firm ignores this externality, there is too little price adjustment from a social perspective.

### Real Rigidity

The firm's incentive to adjust depends on two factors:

1. **Real rigidity:** How far the new optimal price is from the old price. If the optimal price responds little to aggregate output ( $\phi$  small), the gain from adjusting is small even before menu costs.
2. **Curvature of the profit function:** If profits are insensitive to departures from the optimum (flat profit function), the cost of being at the wrong price is small.

Sources of real rigidity and/or profit function insensitivity:

- Labour market frictions
- Countercyclical markups
- Financial market imperfections

## 8. Dynamic New Keynesian Models (Class 9)

*Reading: Romer Ch. 7 intro + 7.1, 7.4, 7.8–7.9*

#### Key features:

- Optimizing households and firms
- Rational expectations
- Imperfect competition, frictions to price adjustment
- No government, no trade, no uncertainty
- Central bank determines the path of the real interest rate

### What changes from the static model?

The static model had one period. Now households and firms make decisions over *infinite time horizons*. Decisions today depend on what agents expect tomorrow. The central bank controls the real interest rate  $r$  — not the money supply  $M$  directly. This is how real-world central banks operate (Fed, ECB, Banxico).

## 9. Households (Dynamic Model)

### Objective Function

$$\sum_{t=0}^{\infty} \beta^t [U(C_t) - V(L_t)], \quad 0 < \beta < 1 \quad (7.1)$$

$$U(C_t) = \frac{C_t^{1-\theta}}{1-\theta}, \quad \theta > 0 \quad (7.2)$$

$$V(L_t) = \frac{B}{\gamma} L_t^\gamma, \quad B > 0, \quad \gamma > 1 \quad (7.3)$$

### What is $\beta^t$ ?

The discount factor. The household values the future less than the present:  $\beta^2 < \beta < 1$ .  $\beta$  close to 1 = patient household.  $\beta$  close to 0 = impatient household.

### Why is $V(L_t) = \frac{B}{\gamma} L_t^\gamma$ constructed this way?

We need a function where working more hurts, and each extra hour hurts more than the last ( $V' > 0$ ,  $V'' > 0$ ). The exponent  $\gamma > 1$  achieves this. The  $\frac{1}{\gamma}$  in front is a normalization so that  $V'(L) = BL^{\gamma-1}$  — clean for the FOC.  $B > 0$  scales the overall level of disutility.

### Are there savings in this model?

Yes — unlike the static model. Each household can consume less today (save) and more tomorrow, or borrow against future income. The Euler equation governs this choice. However, at the *aggregate* level, net saving is zero since there is no government or investment:  $Y_t = C_t$  always holds in aggregate (what one household saves, another borrows).

### Labor Supply FOC

$$V'(L_t) = U'(C_t) \frac{W_t}{P_t} \quad (7.4)$$

In equilibrium  $C_t = L_t = Y_t$ , so:

$$\frac{W_t}{P_t} = \frac{V'(Y_t)}{U'(Y_t)} \quad (7.5)$$

**Substituting functional forms:**

$$V'(L_t) = BL_t^{\gamma-1} = BY_t^{\gamma-1}$$

$$U'(C_t) = C_t^{-\theta} = Y_t^{-\theta}$$

$$BY_t^{\gamma-1} = Y_t^{-\theta} \cdot \frac{W_t}{P_t}$$

$$\boxed{\frac{W_t}{P_t} = BY_t^{\theta+\gamma-1}} \quad (7.6)$$

**Intuition for the labor FOC:**

At the optimum: marginal benefit of working = marginal cost of working.

Marginal benefit =  $U'(C_t) \cdot W_t/P_t$  = extra utility from the extra consumption earned.

Marginal cost =  $V'(L_t)$  = extra disutility from the extra hour worked.

Note:  $C = WL$  would only be true if  $R = 0$  (no profit income). In this model,  $C = Y$  comes from the goods market equilibrium ( $Y = C$ ), not from the budget constraint.

## The Euler Equation and the New Keynesian IS Curve

$$C_t^{-\theta} = (1 + r_t)\beta C_{t+1}^{-\theta}$$

**Where does  $r_t$  come from?**

The household can save or borrow at real interest rate  $r_t$ . Saving \$1 today yields  $1 + r_t$  tomorrow.  $r_t$  is the price of time — how much you are rewarded for waiting. The Euler equation balances the utility of consuming today vs. saving and consuming tomorrow.

**Deriving the New Keynesian IS curve:**

Take logs of the Euler equation:

$$-\theta \ln C_t = \ln(1 + r_t) + \ln \beta - \theta \ln C_{t+1}$$

Divide by  $-\theta$ :

$$\ln C_t = \ln C_{t+1} - \frac{1}{\theta} \ln(1 + r_t) - \frac{1}{\theta} \ln \beta$$

Apply approximations:  $C_t = Y_t$ ,  $\ln(1 + r_t) \approx r_t$ ,  $\beta \approx 1 \Rightarrow \ln \beta \approx 0$ :

$$\boxed{\ln Y_t = \ln Y_{t+1} - \frac{1}{\theta} r_t}$$

**Intuition for the IS curve:**

Output today depends on two things:

- (1)  $\ln Y_{t+1}$ : expected future income. If you expect to be richer tomorrow, you consume more today (*wealth effect*).
- (2)  $r_t$ : the real interest rate. If  $r_t$  rises, saving is more attractive  $\Rightarrow$  consume less today  $\Rightarrow Y_t$  falls.

**Difference from traditional IS:** The traditional IS curve is  $\ln Y_t = -\frac{1}{\theta} r_t$  — no future term. The New Keynesian version adds  $\ln Y_{t+1}$ : what the central bank does *tomorrow* affects the economy *today*.

## 10. Firms' Problem (Dynamic Model)

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**Setup**

Real profits of firm  $i$  at time  $t$ :

$$Y_t \left[ \left( \frac{P_{it}}{P_t} \right)^{1-\eta} - \frac{W_t}{P_t} \left( \frac{P_{it}}{P_t} \right)^{-\eta} \right]$$

Let  $q_t$  = probability that the price set at  $t = 0$  is still in effect at period  $t$ .

**What is  $q_t$ ? Where does it come from?**

This is the **Calvo pricing** mechanism. In each period, each firm receives a random signal allowing it to adjust its price with probability  $1 - \alpha$ , and must keep its old price with probability  $\alpha$ .

$q_t$  is the probability that by period  $t$ , the firm has *not yet* received the signal to adjust since  $t = 0$ . It falls over time: the longer you wait, the more likely you've had a chance to adjust.

This is not triggered by a shock — it is an exogenous friction on the price adjustment process.

**Stochastic discount factor:**

$$\lambda_t \equiv \beta^t \frac{U'(C_t)}{U'(C_0)}$$

**Why not just use  $\beta^t$ ?**

The firm's profits belong to the household. The household does not care about pesos per se — it cares about the utility those pesos provide. A peso of profit in period  $t$  is worth more if the household is consuming little (high marginal utility  $U'(C_t)$ ), and less if it is consuming a lot.

$\lambda_t$  is called “stochastic” because  $C_t$  is not known today — it depends on future shocks. So  $\lambda_t$  is a random variable from today's perspective.

## The Firm's Maximization Problem

$$\max_{P_i} \sum_{t=0}^{\infty} q_t \lambda_t Y_t P_t^{\eta-1} (P_i^{1-\eta} - W_t P_i^{-\eta})$$

### Why no time subscript on $P_i$ ?

Because the firm fixes ONE price today that remains in effect for multiple future periods (with probability  $q_t$ ). That is exactly the nominal rigidity. The firm chooses the single best price for all periods where it will be stuck with it.

### What does the firm take into account when maximizing?

Only its revenues and production costs — NOT the menu cost. The menu cost enters separately as a fixed cost paid once if the firm decides to adjust. The firm first finds the optimal  $P_i^*$  ignoring the menu cost, then compares  $\Pi_{adj} - \Pi_{fixed}$  against the menu cost to decide whether to adjust.

## Simplification

Using: (i)  $\lambda_t Y_t P_t^{\eta-1}$  varies negligibly, and (ii) second-order approximation of  $F$ :

$$F(P_i, P_t^*) \simeq F(P_t^*, P_t^*) - K(P_i - P_t^*)^2, \quad K > 0$$

### Why is $K$ a constant and not a function of $(P_i - P_t^*)$ ?

$K$  is the second derivative of  $F$  evaluated at  $P_t^*$  — it depends on parameters like  $\eta$  and on  $P_t^*$ . But since the economy stays close to the flexible-price equilibrium,  $P_t^*$  does not vary much, so  $K$  is approximately constant. It is a positive scalar that scales how costly it is to be away from the optimum — not a function of the distance itself.

### Deriving the simplified problem:

Substitute the approximation of  $F$ :

$$\max_{P_i} \sum_{t=0}^{\infty} q_t \lambda_t Y_t P_t^{\eta-1} [F(P_t^*, P_t^*) - K(P_i - P_t^*)^2]$$

Split the sum. The first part does not depend on  $P_i$  — drop it. Treat  $\lambda_t Y_t P_t^{\eta-1}$  and  $K$  as constants and drop them (they do not affect the location of the optimum). Maximizing the negative term is equivalent to minimizing:

$$\min_{P_i} \sum_{t=0}^{\infty} q_t (P_i - P_t^*)^2$$

## FOC and Solution

**Taking the FOC:**

Differentiate with respect to  $P_i$  (which does NOT depend on  $t$ , so it comes outside the sum):

$$\frac{\partial}{\partial P_i} \sum_{t=0}^{\infty} q_t (P_i - P_t^*)^2 = \sum_{t=0}^{\infty} q_t \cdot 2(P_i - P_t^*) = 0$$

$$P_i \sum_{t=0}^{\infty} q_t = \sum_{t=0}^{\infty} q_t P_t^*$$

$$P_i = \sum_{t=0}^{\infty} \frac{q_t}{\sum_{\tau} q_{\tau}} P_t^* = \sum_{t=0}^{\infty} \omega_t P_t^*$$

**Why does  $P_i$  come outside the sum?**

The derivative passes inside the sum because of linearity. Then  $P_i$  comes out because it does NOT depend on the summation index  $t$  — it is the same price in every period. Only  $q_t$  and  $P_t^*$  depend on  $t$  and remain inside.

The price set today is a **weighted average** of all future optimal prices  $P_t^*$ , weighted by  $\omega_t = q_t / \sum_{\tau} q_{\tau}$  (the probability that the price is still in effect, normalized to sum to 1).

With uncertainty:

$$P_i = \sum_{t=0}^{\infty} \omega_t \mathbb{E}_0(P_t^*)$$

**The New Keynesian Aggregate Supply Curve****Expressing  $p_t^*$  in terms of  $m_t$  and  $p_t$ :**

We know  $P_t^*/P_t = \frac{\eta}{\eta-1} \cdot W_t/P_t$  and from the household's condition:  $W_t/P_t = B Y_t^{\theta+\gamma-1}$ . In logs, choosing  $B$  so that the constant  $c = \ln \frac{\eta}{\eta-1} + \ln B = 0$ :

$$p_t^* = p_t + \phi y_t, \quad \phi \equiv \theta + \gamma - 1$$

Using  $y_t = m_t - p_t$  (aggregate demand in logs):

$$p_t^* = p_t + \phi(m_t - p_t) = \phi m_t + (1 - \phi)p_t$$

**The New Keynesian Aggregate Supply curve:**

$$p_i = \sum_{t=0}^{\infty} \omega_t \mathbb{E}_0 \left( \phi m_t + (1 - \phi)p_t \right)$$

The price set today is a weighted average of expected future values of money  $m_t$  and the price level  $p_t$ , across all periods where this price will be in effect.



## 11. Summary and Connection

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### **The thread connecting the static and dynamic models:**

Both models reach the same essential conclusion: the firm's optimal price depends on the general price level  $p_t$  and output  $y_t$ . With nominal rigidity, firms cannot always adjust to that optimum.

**Static model (Class 8):** One period, menu costs, firms compare  $\Pi_{adj} - \Pi_{fixed}$  against the cost of adjusting.

**Dynamic model (Class 9):** Infinite horizon, Calvo pricing ( $q_t$ ), firms set a price today that is the weighted average of all future optimal prices.

**What comes next — Calvo model:** Formalizes  $q_t$  explicitly. In each period, each firm can adjust with probability  $1 - \alpha$  and cannot with probability  $\alpha$ . Aggregating across firms yields the **New Keynesian Phillips Curve**:

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa y_t$$

which relates today's inflation to expected future inflation and the output gap.